

Standard Process Inc.

1200 W. Royal Lee Drive
Palmyra, WI 53156

ANALYTICAL REPORT

Reported On: 17Oct2023

Eurofins Sample Code: 468-2023-03310106
Client Sample Code: HEMP OIL COMPLEX
Sample Description: HEMP OIL COMPLEX
5751-3-* -1-1

Industrial Hemp Cannabinoid Profile	Reference	Accreditation	Completed	Sub
	AOAC 2018.11 (modified)		11Apr2023	3

Parameter	Result
Total THC (THC + (THCA x 0.877))	0.23655 mg/capsule
Cannabichromene (CBC)	0.25194 mg/capsule
Cannabichromenic Acid (CBCA)	0.21717 mg/capsule
Cannabicyclol (CBL)	<0.01425 mg/capsule
Cannabidiol (CBD)	2.9127 mg/capsule
Cannabidiolic Acid (CBDA)	4.788 mg/capsule
Cannabidivarin (CBDV)	<0.00456 mg/capsule
Cannabidivarinic Acid (CBDVA)	0.020805 mg/capsule
Cannabigerol (CBG)	0.06555 mg/capsule
Cannabigerolic Acid (CBGA)	0.026619 mg/capsule
Cannabinol (CBN)	0.018981 mg/capsule
Cannabinolic Acid (CBNA)	<0.00228 mg/capsule
Delta 8-THC	<0.00969 mg/capsule
Delta 9-THC	0.17499 mg/capsule
Tetrahydrocannabinolic acid (THCA)	0.07011 mg/capsule
Tetrahydrocannabivarin (THCV)	<0.00456 mg/capsule
Tetrahydrocannabivarinic Acid (THCVA)	<0.00228 mg/capsule
Total Cannabinoids	8.55 mg/capsule
Total CBD (CBD + (CBDA x 0.877))	7.125 mg/capsule

The Hemp Oil Complex (Lot#5751-3-* -1-1) meets the requirements of <0.3% for Delta 9 THC Levels.

M904G - Glucoraphanin	Reference	Accreditation	Completed	Sub
	Internal method		11Apr2023	4

Parameter	Result
Glucoraphanin	6.02 mg/g

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**Pesticides <USP 561> (LC-MS/
MS and GC-MS/MS)**

Reference
USP 561

Accreditation
ISO/IEC 17025:2017
A2LA 2993.01

Completed
04Apr2023

Parameter	Result
Acephate	<0.1 mg/kg
Alachlor	<0.05 mg/kg
Aldrin and dieldrin (sum of)	<0.05 mg/kg
Azinphos-ethyl	<0.1 mg/kg
Azinphos-methyl	<1 mg/kg
Bromophos-ethyl	<0.05 mg/kg
Bromophos-methyl	<0.05 mg/kg
Bromopropylate	<3 mg/kg
Chlordane (sum)	<0.05 mg/kg
Chlorfenvinphos	<0.5 mg/kg
Chlorpyrifos	<0.2 mg/kg
Chlorpyrifos-methyl	<0.1 mg/kg
Chlorthal-dimethyl	<0.01 mg/kg
Cyfluthrin I-IV	<0.1 mg/kg
Cyhalothrin, lambda-	<1 mg/kg
Cypermethrin I-IV	<1 mg/kg
DDT (sum)	<1.0 mg/kg
Deltamethrin	<0.5 mg/kg
Diazinon	<0.5 mg/kg
Dichlofluanid	<0.1 mg/kg
Dichlorvos	<1 mg/kg
Dicofol, o,p' & p,p'-	<0.5 mg/kg
Dimethoate and omethoate (sum of)	<0.1 mg/kg
Endosulfan (sum)	<3.0 mg/kg
Endrin	<0.05 mg/kg
Ethion	<2 mg/kg
Etrinfos	<0.05 mg/kg
Fenchlorophos (sum)	<0.1 mg/kg
Fenitrothion	<0.5 mg/kg
Fenpropathrin	<0.03 mg/kg
Fensulfothion (sum)	<0.05 mg/kg
Fenthion (sum)	<0.05 mg/kg
Fenvalerate I&II	<1.5 mg/kg
Flucythrinate I&II	<0.05 mg/kg

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Reference
USP 561

Accreditation
ISO/IEC 17025:2017
A2LA 2993.01

Completed
04Apr2023

Parameter	Result
Fluvalinate, tau- I&II	<0.05 mg/kg
Fonofos	<0.05 mg/kg
Heptachlor (sum)	<0.05 mg/kg
Hexachlorobenzene (HCB)	<0.1 mg/kg
Hexachlorocyclohexane isomers (other than gamma)	<0.3 mg/kg
Lindane (gamma-HCH)	<0.6 mg/kg
Malathion and malaoxon (sum of)	<1 mg/kg
Mecarbam	<0.05 mg/kg
Methacrifos	<0.05 mg/kg
Methamidophos	<0.05 mg/kg
Methidathion	<0.2 mg/kg
Methoxychlor	<0.05 mg/kg
Mirex	<0.01 mg/kg
Monocrotophos	<0.1 mg/kg
Parathion-ethyl and paraoxon-ethyl (sum of)	<0.5 mg/kg
Parathion-methyl and paraoxon-methyl (sum of)	<0.2 mg/kg
Pendimethalin	<0.1 mg/kg
Pentachloranisole	<0.01 mg/kg
Permethrin (sum of isomers)	<1 mg/kg
Phosalone	<0.1 mg/kg
Phosmet	<0.05 mg/kg
Piperonyl butoxide	<3 mg/kg
Pirimiphos-ethyl	<0.05 mg/kg
Pirimiphos-methyl (sum)	<4 mg/kg
Procymidone	<0.1 mg/kg
Profenofos	<0.1 mg/kg
Prothiofos	<0.05 mg/kg
Pyrethrum (total isomers)	<3 mg/kg
Quinalphos	<0.05 mg/kg
Quintozene (sum)	<1 mg/kg
S 421	<0.02 mg/kg
Tecnazene	<0.05 mg/kg
Tetradifon	<0.3 mg/kg

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Pesticides <USP 561> (LC-MS/MS and GC-MS/MS)	Reference USP 561	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 04Apr2023
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Parameter	Result
Vinclozolin	<0.4 mg/kg
The Hemp Oil Complex (Lot#5751-3-* -1-1) meets the requirements for pesticides as defined by USP 561.	

Product Specification Review - Pesticides	Reference Internal method	Completed 11Apr2023
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Parameter	Result
Conformance to product specification	Completed

Heavy Metals As-Cd-Pb-Hg (Low LOQ, ICP-MS)	Reference AOAC 2013.06	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 03Apr2023
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Parameter	Result
Arsenic (As)	0.31 ppm
Cadmium (Cd)	0.0059 ppm
Lead (Pb)	0.013 ppm
Mercury (Hg)	<0.0020 ppm
The Hemp Oil Complex (Lot#5751-3-* -1-1) meets the requirements of heavy metals as defined by USP 2232.	

Fatty Acids-Omega 6 & 3 %W/W (AOCS, Marine Oils)	Reference AOCS Ce 1b-89 mod.	Accreditation	Completed 06Apr2023	Sub Fatty
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Parameter	Result
Fatty Acid Profile, % Weight	Reported as Fatty Acids
C08:0 Octanoic (Caprylic)	<0.02 %
C10:0 Decanoic (Capric)	<0.02 %
C11:0 Undecanoic (Hendecanoic)	<0.02 %
C12:0 Dodecanoic (Lauric)	<0.02 %
C14:0 Tetradecanoic (Myristic)	0.9 %
C14:1 Tetradecenoic (Myristoleic)	<0.02 %
C15:0 Pentadecanoic	0.1 %
C15:1 Pentadecenoic	<0.02 %
C16:0 Hexadecanoic (Palmitic)	5.9 %
C16:1 Hexadecenoic (Palmitoleic)	1.5 %
C16:2 Hexadecadienoic	0.1 %
C16:3 Hexadecatrienoic	0.04 %
C16:4 Hexadecatetraenoic	0.1 %
C17:0 Heptadecanoic (Margaric)	0.2 %
C17:1 Heptadecenoic (Margaroleic)	0.1 %
C18:0 Octadecanoic (Stearic)	1.2 %

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Fatty Acids-Omega 6 & 3 %W/W (AOCS, Marine Oils)	Reference AOCS Ce 1b-89 mod.	Accreditation	Completed 06Apr2023	Sub 1
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Parameter	Result
C18:1 Octadecenoic (Oleic)	9.7 %
C18:2 Octadecadienoic (Linoleic + isomers)	2.0 %
C18:2 Octadecadienoic Omega 6 (Linoleic)	1.7 %
C18:3 Octadecatrienoic (Linolenic + isomers)	0.5 %
C18:3 Octadecatrienoic Omega 3 (Alpha Linolenic)	0.4 %
C18:3 Octadecatrienoic Omega 6 (Gamma Linolenic)	0.1 %
C18:4 Octadecatetraenoic	0.6 %
C18:4 Octadecatetraenoic Omega 3 (Stearidonic)	0.6 %
C20:0 Eicosanoic (Arachidic)	0.2 %
C20:1 Eicosenoic (Gondoic + isomers)	4.9 %
C20:2 Eicosadienoic	0.5 %
C20:2 Eicosadienoic Omega 6	0.5 %
C20:3 Eicosatrienoic	0.2 %
C20:3 Eicosatrienoic Omega 3	0.1 %
C20:3 Eicosatrienoic Omega 6	0.1 %
C20:4 Eicosatetraenoic (Arachidonic + isomers)	1.3 %
C20:4 Eicosatetraenoic Omega 3	0.6 %
C20:4 Eicosatetraenoic Omega 6 (Arachidonic)	0.7 %
C20:5 Eicosapentaenoic	6.4 %
C20:5 Eicosapentaenoic Omega 3	6.4 %
C21:5 Heneicosapentaenoic	0.3 %
C21:5 Heneicosapentaenoic Omega 3	0.3 %
C22:0 Docosanoic (Behenic)	0.1 %
C22:1 Docosenoic (Erucic + isomers)	4.5 %
C22:2 Docosadienoic	0.1 %
C22:2 Docosadienoic Omega 6	0.1 %
C22:3 Docosatrienoic	<0.02 %
C22:3 Docosatrienoic, Omega 3	<0.02 %
C22:4 Docosatetraenoic	0.2 %
C22:4 Docosatetraenoic Omega 6	0.2 %
C22:5 Docosapentaenoic	1.3 %

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Fatty Acids-Omega 6 & 3 %W/W (AOCS, Marine Oils)	Reference AOCS Ce 1b-89 mod.	Accreditation	Completed 06Apr2023	Sub 1
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Parameter	Result
C22:5 Docosapentaenoic Omega 3	1.0 %
C22:5 Docosapentaenoic Omega 6	0.3 %
C22:6 Docosahexaenoic	12.8 %
C22:6 Docosahexaenoic Omega 3	12.8 %
C24:0 Tetracosanoic (Lignoceric)	0.2 %
C24:1 Tetracosenoic (Nervonic)	0.7 %
Sum of Omega 3 Isomers	22.2 %
Sum of Omega 6 Isomers	3.6 %
Total Fatty Acids Calc.	56.5 %
Total Fat as Triglycerides	58.9 %
Total Monounsaturated Fatty Acids	21.4 %
Total Polyunsaturated Fatty Acids	26.2 %
Total Saturated Fatty Acids	8.8 %

Moulds - FDA BAM Chapter 18 mod.	Reference FDA BAM Chapter 18 mod.	Accreditation	Completed 10Apr2023	Sub
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Parameter	Result
Mold	< 10 cfu/g
Yeast	< 10 cfu/g

Aerobic Plate Count - AOAC 966.23	Reference AOAC 966.23	Accreditation	Completed 10Apr2023	Sub 2
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Parameter	Result
Aerobic Plate Count	< 10 cfu/g

Escherichia coli - BAM Chapter 4	Reference FDA BAM Chapter 4	Accreditation	Completed 10Apr2023	Sub 2
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Parameter	Result
E. coli	< 3 MPN/g

The Hemp Oil Complex (Lot#5751-3-* -1-1) meets the requirements for microbiological levels as defined by USP 2023 (Botanical Extracts).

Comments:

Rev.01: Units updated per client request.

Subcontracting partners:

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- 1 - Eurofins Nutrition Analysis Center, Iowa
 - 2 - Eurofins Microbiology Laboratories (Los Angeles), California
 - 3 - Eurofins Food Chemistry Testing US Madison, WI
 - 4 - Eurofins Botanical Testing US Brea, California

Respectfully Submitted,



A handwritten signature in black ink that reads 'D. Walkenhorst'.

Dana Walkenhorst
Analytical Services Manager

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